

SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)

Assessment Tool Weightage: 50%

Duration: 1 Hour

QUESTIONS: 5 Total Marks:50

Annexure A

Scenario Based Assignment

Code of Conduct:

I SANJAI S(192511018)(Name / Reg No) certify that this submission is my original work

and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:harsha

Scenario Based Assignment – Artificial Intelligence 1.

AI-Powered Drone for Emergency Medicine Delivery

i. Greedy Best-First Search (GBFS)

Greedy Best-First Search selects the path that appears closest to the destination based only on the heuristic value ($h(n)$), such as straight-line distance. It ignores the actual travel cost.

Strengths:

- Fast decision-making.
- Requires less computation.
- Suitable for emergency situations where quick responses are required.

Weaknesses:

- Does not guarantee the shortest or safest path.
- May choose blocked or risky routes.
- Can get trapped in dead ends because it ignores actual path cost.

ii. A Search*

A* Search uses the evaluation function:

$$f(n) = g(n) + h(n)$$

Where:

- $g(n)$ = Actual cost from the start.
- $h(n)$ = Estimated cost to the goal.

A* balances both the travel cost and estimated distance, helping the drone choose safer and more efficient routes.

Advantages:

- Finds the optimal path when the heuristic is admissible.
- Avoids unnecessarily expensive routes.
- Adapts better to changing conditions.

iii. Impact of Heuristic Accuracy

- **Accurate heuristic:** Faster search and better route selection.
- **Overestimated heuristic:** A* may lose optimality.

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Poor heuristic: Both algorithms explore more unnecessary nodes.

Greedy Best-First Search depends entirely on heuristic accuracy, whereas A* is less affected because it also considers actual cost.

iv. Effect of Dynamic Changes

Blocked roads and changing weather require replanning.

Greedy Best-First Search

- Frequently chooses blocked routes.
- Needs repeated searching.

A*

- Recalculates costs efficiently.
- Produces better alternative routes.
- More reliable under changing conditions.

v. Recommended Algorithm

*Recommended: A Search**

Justification:

- Produces optimal routes.
- Considers safety and travel cost.
- Performs better in dynamic environments.
- Provides reliable decisions for emergency medicine delivery.

2. Smart City Traffic Signal Optimization

Problem Formulation This is

an optimization problem.

Objective:

- Minimize waiting time.
- Minimize fuel consumption.
- Reduce traffic congestion.

Traditional search algorithms are inefficient because:

- The search space is extremely large.
- Traffic changes continuously.
- Exhaustive search requires excessive computation.

i. Hill Climbing

Hill Climbing starts with an initial traffic signal configuration and continuously selects a neighboring solution with better performance.

Advantages

- Simple implementation.
- Fast execution.
- Low memory usage.

Limitations

- Can become trapped in local optimum.
- Cannot guarantee the best traffic schedule.

ii. Local Maxima, Plateaus and Ridges

Local Maximum

- AI believes traffic flow is optimal although a better solution exists.

Plateau

- Multiple signal timings produce nearly identical performance.
- No clear improvement direction.

Ridge

- Improvement requires several coordinated signal changes.
- Hill Climbing struggles to move toward the better solution.

iii. Simulated Annealing and Genetic Algorithms

Simulated Annealing

- Occasionally accepts worse solutions.
- Escapes local maxima.
- Finds better global solutions.

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Genetic Algorithm

Uses population-based search.

- Applies selection, crossover and mutation.
- Explores many solutions simultaneously.
- Highly suitable for large optimization problems.

iv. Recommended Approach

Recommended: Genetic Algorithm

Justification

- Highly scalable.
 - Handles changing traffic efficiently.
 - Avoids local optima.
 - Produces near-optimal signal timings.
- ## **3. Autonomous Mars Rover**

i. Why Online Search?

The rover has no complete map before starting.

Online Search:

- Makes decisions while exploring.
- Learns continuously.
- Updates routes according to new information.

Offline search is unsuitable because it requires complete environmental knowledge before movement.

ii. Challenges

Partial Observability

- Unknown obstacles.
- Limited sensor range.

Dynamic Environment •

Dust storms.

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- Slippery terrain.
- Unexpected rocks.

These conditions require continuous adaptation.

iii. Internal Model The rover continuously updates:

- Obstacle locations.
- Safe paths.
- Visited regions.
- Sample locations.
- Terrain information.

This internal map improves future decisions.

iv. Comparison

Online Search	Uninformed Search	Informed Search
Learns while moving	No heuristic	Uses heuristic
Suitable for unknown maps	Explores blindly	Needs heuristic
Highly adaptive	Slow	Faster

v. Recommended Approach

Recommended: Online Search Agent

Reasons

- Works in unknown environments.
- Learns continuously.
- Safely avoids hazards.
- Adapts to unexpected situations.

4. University Exam Timetable (Constraint Satisfaction Problem)

Problem Formulation

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Variables

Exam subjects

- Time slots
- Exam halls

Domains

Possible values for:

- Available time slots
- Available rooms

Constraints

- No overlapping exams.
- Limited hall capacity.
- Faculty availability.
- Subject precedence.
- Special accommodations.

i. Variables, Domains and Constraints

Variables represent exams.

Domains represent possible time slots and halls. Constraints ensure valid scheduling **ii.**

Backtracking Search

Steps:

1. Assign a subject to a time slot.
2. Check constraints.
3. If constraints fail, undo assignment.
4. Try another assignment.
5. Continue until all exams are scheduled.

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iii. Forward Checking

Forward Checking:

- Removes invalid choices early.
Detects conflicts before deeper search.
- Reduces unnecessary computation.
- Speeds up timetable generation.

iv. Real-world Challenges Challenges:

- Thousands of students.
- Hundreds of subjects.
- Faculty scheduling.
- Hall shortages.
- Continuous timetable modifications.

Best Strategy

Backtracking + Forward Checking + Constraint Propagation

This combination improves scalability and efficiency.

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v. Recommended Strategy

Recommended: Minimax + Alpha-Beta Pruning + Evaluation Function

Justification

- Produces near-optimal decisions.
- Reduces computation.
- Meets real-time constraints.
- Performs efficiently in complex strategy games.

Conclusion

Artificial Intelligence algorithms solve different real-world problems based on their strengths.

- A* Search is ideal for dynamic pathfinding, Genetic Algorithms excel in optimization, Online Search Agents handle uncertain environments, CSP efficiently solves scheduling problems, and Minimax with Alpha-Beta Pruning enables intelligent real-time game decision-making.